

ECON 21030: Honors Econometrics

Applications of Some Results from Linear Algebra

Spring Quarter 2024

For k -dimensional random vector X with mean μ and covariance matrix Σ and k -by- k matrix A , we have:

$$E[X'AX] = \mu' A \mu + \text{trace}(A\Sigma)$$

This result follows from the fact that because $X'AX$ is a scalar, $X'AX = \text{trace}(X'AX)$. Thus:

$$\begin{aligned} E[\text{trace}(X'AX)] &= E[\text{trace}(AXX')] \\ &= \text{trace}(E[AXX']) \\ &= \text{trace}(AE[XX']) \end{aligned}$$

Now $\Sigma = E[XX'] - \mu\mu'$, so we have:

$$\begin{aligned} \text{trace}(AE[XX']) &= \text{trace}(A\Sigma + A\mu\mu') \\ &= \text{trace}(A\Sigma) + \text{trace}(A\mu\mu') \\ &= \text{trace}(A\Sigma) + \text{trace}(\mu' A \mu) \\ &= \mu' A \mu + \text{trace}(A\Sigma) \end{aligned}$$

where the last line follows from the fact that $\mu' A \mu$ is a scalar, and hence equal to its trace.

A fact we used repeatedly in this proof is that for n -by- m and m -by- n matrices A and B , $\text{trace}(AB) = \text{trace}(BA)$. This can be seen directly from the fact that $\text{trace}(AB)$ is the sum of the diagonal elements of AB , or:

$$\begin{aligned} \text{trace}(AB) &= \sum_{i=1}^n AB_{ii} \\ &= \sum_{i=1}^n \sum_{j=1}^m A_{ij} B_{ji} \\ &= \sum_{j=1}^m \sum_{i=1}^n B_{ji} A_{ij} \\ &= \text{trace}(BA) \end{aligned}$$

For $X \sim N(\mu, \Sigma)$ and matrix A of appropriate dimension, AX is also normal because normals are closed under linear transformations. Its mean is $A\mu$ (this should be obvious). The variance

is:

$$\begin{aligned} \text{Var}(AX) &= E[AXX'A] - A\mu\mu'A' \\ &= E[A(XX' - \mu\mu')A'] \\ &= A\Sigma A' \end{aligned}$$

Recall that a real k -by- k matrix A has a [Cholesky decomposition](#) whenever there is a lower-triangular L with strictly positive diagonal entries and such that $A = LL'$. You can see that Cholesky is a type of [LU decomposition](#) where $U = L'$, if you're more familiar with that idea. Square matrices have a Cholesky decomposition if and only if they are [positive definite](#). Positive definite means that both $x'Ax > 0$ for all $x \neq 0$ and that A is [symmetric](#), which means it satisfies $A = A'$. Because the Cholesky factors V are triangular with strictly positive diagonal elements they are also invertible.

Variance-covariance matrices Σ are symmetric by definition. And they are positive definite because if k -dimensional random vector X has $\text{Var}(X) = \Sigma$, $\text{Var}(cX) = c\Sigma c'$ for k -dimensional vector c . But if cX is itself a random variable, it must have positive variance.¹ Therefore $c\Sigma c' > 0$.

As a result, Σ always has a factorization $\Sigma = VV'$, for some V invertible, whenever it is a well-defined variance covariance matrix. This implies $V^{-1}(X - \mu) \sim N(0, I)$, i.e., independent standard normal. This follows from the fact $V^{-1}(X - \mu)$ is normal with mean zero, while its variance satisfies:

$$V^{-1}\Sigma V^{-1'} = V^{-1}VV'V'^{-1} = I$$

We have discussed how for a k -dimensional standard normal vector X , $X'X \sim \chi^2(k)$ (because it is the sum of k independent standard normals). This implies that for $X \sim N(\mu, \Sigma)$, $(X - \mu)' \Sigma^{-1} (X - \mu) \sim \chi^2(k)$. The proof of this is an extension of the previous result, since we have:

$$\begin{aligned} (X - \mu)' \Sigma^{-1} (X - \mu) &= (X - \mu)' (VV')^{-1} (X - \mu) \\ &= (X - \mu)' V'^{-1} V^{-1} (X - \mu) \\ &= (V^{-1}(X - \mu))' (V^{-1}(X - \mu)) \\ &= Y'Y \end{aligned}$$

where $Y \sim N(0, I)$.

Another special case deals with matrices that are [orthogonal](#), which applies to A if it satisfies $AA' = I$. For any orthogonal matrix A and $X \sim N(0, I)$, we have $Y = AX \sim N(0, I)$. This follows immediately from the above because $AA' = I$.

Symmetric matrices can be decomposed into $A = PDP'$, where P is the orthogonal matrix of eigenvectors and D is the diagonal matrix of the eigenvalues of A . If a symmetric matrix is also [idempotent](#) (meaning $AA = A$), then its eigenvalues are either zero or one. This comes from the fact $A = AA = PDP'PDP' = PDDP'$, so $D^n = D$ for all $n > 1$, which requires that diagonal entries to be either zero or one.

¹Except in degenerate cases where e.g., $c = 0$ or the elements of X are perfectly correlated.

For a standard normal $X \sim N(0, I)$ and symmetric idempotent A , it turns out that $X'AX \sim \chi^2(\text{trace}(A))$. This follows because:

$$\begin{aligned} X'AX &= X'PDP'X = (P'X)'DP'X \\ &= Y'DY \end{aligned}$$

where $Y \sim N(0, I)$ by the result about orthogonal matrices established above. Because D is diagonal, $Y'DY$ is just the sum of k squared standard normals multiplied by the eigenvalues of D , which as established earlier are either zero or one. Because $\text{trace}(A)$ is equal to the sum of its eigenvalues, there are exactly $\text{trace}(A)$ eigenvalues equal to one, and hence $Y'DY \sim \chi^2(\text{trace}(A))$.